Q03: Why RL After Pre-Training?

Paradise (nuc13)

October 2026

The problem likelihood cannot solve

- Pre-training: next-token likelihood on a corpus of *how people write*
- Assistant quality: *comparisons* between whole responses
- "Better" is not a likelihood — reward is an arbitrary scalar over full outputs [**?**]

The RLHF recipe (InstructGPT)

1. Collect pairwise preference rankings
2. Fit reward model r_ϕ (Bradley–Terry)
3. Optimize policy: reward $- \beta$ KL to reference [**?**]

The RLHF recipe (InstructGPT)

1. Collect pairwise preference rankings
2. Fit reward model r_ϕ (Bradley–Terry)
3. Optimize policy: reward $- \beta$ KL to reference [**?**]

PPO [**?**]: LLM = policy, tokens = actions, value net for advantages

Why the KL anchor matters

- Reward model is valid only near its training distribution
- Unanchored: reward hacking + degenerate collapse
- KL = **trust region** for the reward model, not a safety feature

DPO: the closed-form shortcut

- RLHF optimum: $\pi^*(y|x) \propto \pi_{\text{ref}} \exp(r/\beta)$
- Solve for r, substitute into Bradley–Terry: $Z(x)$ cancels
- Pure classification loss on log-ratios — no reward model, no sampling [**?**]
- Trade-off: no reusable reward; distribution-shift sensitivity

GRPO: drop the value network

- Sample G responses per prompt; score all
- Advantage $= (r_i - \text{mean})/\text{std}$ over the group [**?**]
- Shines with verifiable rewards (tests, exact answers)
- Backbone of modern reasoning-model RL

Shared failure modes

- Reward hacking (learned r_ϕ has exploits)
- Length bias (longer looks better)
- Evaluator gaming (judge blind spots)
- Preferences $\neq$ truth (raters, not reality)

Summary

RL enters because "better" is a comparison, not a likelihood. RLHF: learned reward + KL trust region. DPO: closed form, no machinery. GRPO: group baselines, no value net. All three: optimize for raters — with the risks that implies.